

## 4.1 Multimodal Foundations

Five jobs: represent, align, fuse, translate, co-learn. CLIP scores; concat fuses.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

Module 2 leaves pure text. **Multimodal** systems see more than one stream: image and caption, video and audio, a PDF's pixels and its words. The five jobs are **representation, alignment, fusion, translation, co-learning** (Baltrušaitis et al.). Complementary signal (the label says flammable; the photo does not) and redundant signal (lips plus audio).

If you only train on text, you cannot answer "what is in this photo?" without an image tower or a captioner. That is why Weeks 4–5 exist.

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**First time this method appears.** Multimodal learning is more than "glue an image to a caption."

**What.** Five jobs (Baltrušaitis et al.): represent, align, fuse, translate, co-learn. A model can score a pair (CLIP) without decoding a sentence. **Why.** Pixels, text, and audio are different sensors. Concatenation is one fusion; it is not the only job. **Architecture.** Separate encoders per modality, then a fuse (concat, add, or cross-attention) or a score (cosine). **How.** Pick the job first. Retrieval needs a score. Captioning needs a decoder. Classification can be a linear head on a frozen encoder. **Formula.** Coordinated:  $s(i, t) = \cos(f(i), g(t))$ . Joint concat:  $[f(i); g(t)] W$ . **Tradeoffs.** + You can reuse a frozen encoder. – Concat grows width; alignment needs paired data; "multimodal" without naming the job is not a project.

### 1. Five jobs, not one architecture

Five jobs. CLIP scores; captioning translates; concat fuses.



| Job       | Question                           | This course                                |
|-----------|------------------------------------|--------------------------------------------|
| Represent | One vector, or two towers?         | Joint vs coordinated                       |
| Align     | Which region matches which phrase? | Attention, or one cosine on the whole pair |
| Fuse      | When do you mix features?          | Early / late / cross-attention             |
| Translate | Map one stream into another?       | Captioning (BLIP), not CLIP                |
| Co-learn  | Can a rich stream help a poor one? | Zero-shot in 4.3                           |

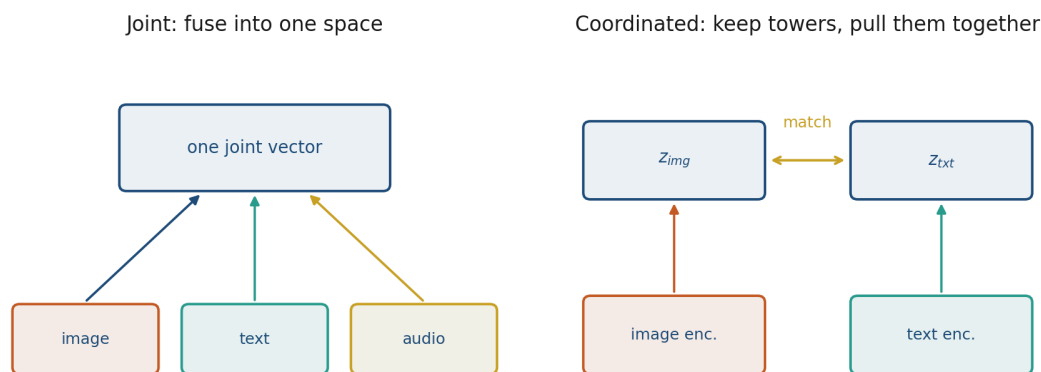
Translation is not fusion. Captioning **maps** image  $\rightarrow$  text. CLIP **scores** image  $\leftrightarrow$  text. If your demo must output a sentence, you need a decoder. If it must search a gallery, you need two towers and a cosine.

A detector that puts a box on “the red mug” is alignment at **region** level. A single cosine on the whole image vs the whole caption is alignment at **instance** level. Know which one your project needs.

## 2. Joint versus coordinated

**Joint:** mash the streams into **one** vector (early fusion, a concatenating MLP). Missing a modality at test time hurts.

**Coordinated:** keep a tower per modality, then pull the towers together with a similarity (CLIP is this; canonical correlation analysis is another coordinated family). You can query with either side.



On the board: two arrows that you **add** (joint) vs two arrows that you **dot** (coordinated). CLIP never concatenates pixels with token ids during pretraining; it only scores pairs.

We do not derive Deep CCA. The slogan is enough: a coordination **loss**, not a concatenating layer.

## 3. Teaching this note

~**16 minutes**. Five-job table, then joint vs coordinated with the 2-D cosine example. Play Yannic Kilcher CLIP **4:40–14:40** (two towers as a coordinated space). Zero-shot details wait for 4.3; captioning waits for BLIP.

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## 4. Worked example

Image vector  $v = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ . Matching caption  $t_+ = \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix}$ , other caption  $t_- = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ .

$$\cos(v, t_+) = 0.8 / \sqrt{0.68} \approx 0.97, \quad \cos(v, t_-) = 0.$$

Coordinated retrieval ranks by cosine. Joint concat needs both streams.

| pair     | vectors                      | cosine |
|----------|------------------------------|--------|
| match    | $v = [1, 0], t = [0.8, 0.2]$ | 0.97   |
| mismatch | $v = [1, 0], t = [0, 1]$     | 0      |

Coordinated retrieval: rank captions by cosine. Joint fusion:  $\text{concat}(v, t)$  is 4-D; you need **both** at test time, and a typed query with no image has the wrong shape.

If the image is missing, a joint MLP trained on concatenations cannot score the text alone unless you trained a special “image dropout” path. A coordinated text tower still embeds the query.

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## 5. Where students get stuck

- Concatenating then calling it CLIP.
  - Assuming a joint model can take a text-only query at test time.
  - Mixing alignment (what matches what) with fusion (when you mix features) with translation (when you generate the other stream).
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## 6. Video

Watch [Yannic Kilcher: OpenAI CLIP, Connecting Text and Images](#).

Play **4:40–14:40**. Pause when the two encoders emit vectors in one space, and when a cosine (not a concatenating MLP) is the interaction. That is coordinated representation.

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## 7. Practice

1. Your app must search photos with a typed query. Joint or coordinated? Why?
2. Give one example of alignment that is not “the whole image vs the whole caption.”

3. What breaks if a joint model never saw audio-only examples and you drop the image at test time?
4. Compute  $\cos\left(\begin{bmatrix} 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 4 \\ 3 \end{bmatrix}\right)$ . If you fuse by concatenation instead, what is the dimension of  $\text{concat}(u, v)$ ?
5. Two images  $v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ ,  $v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$  and one query  $q = \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$ . Which image ranks first by cosine?
6. Captioning versus CLIP, in one sentence each: which job from the five-job table?
7. Co-learning: English captions are plentiful, labels in another language are not. What is the shared stream that lets the rich side help the poor side?